Resonance Warp in PGA–DST: Layer-Hopping vs On-Demand Phase Entry

https://github.com/hypernumbernet

August 8, 2026

Abstract

We develop a hypothetical interstellar propulsion framework within the projective geometric algebra (PGA) extension of dual spacetime theory (DST). Gravity and inertia emerge from torsional mismatch between particle-intrinsic usual and dual spacetimes, lifted into $G(3,1,1)$ with a ten-dimensional bivector generator space (hyperbolic, cyclic, and null sectors). The extended motor $M = \exp(\Omega^{(5)}_{\mathrm{biv}}) = RT$ is unitary ($M\widetilde{M} = 1$), preserving causal propagation on the light cone $\eta_{\mu\nu}\lambda^\mu\lambda^\nu = 0$. The torsion scalar J changes sign at odd multiples of π in the dual rotor angle θ_a, yielding repulsive layers ($J < 0$) that enable *effective* superluminal coasting without local causality violation.

Unlike Alcubierre-type metric engineering, warp in PGA–DST is not about bending a spacetime fabric with exotic matter; it is about coherently tuning collective dual-rotor resonance and entering $J < 0$ layers via circularly polarised THz driving. We quantitatively compare two operational regimes: (1) *layer-hopping*, exploiting slow cosmic phase drift of background dual rotors, and (2) *on-demand entry*, applying active phase driving. For million-tonne-class vessels at 0.1–0.5c, layer-hopping reduces entry energy by 4–7 orders of magnitude, into the kWh–low MWh range. All derivations respect the $G(3,1,1)$ algebra, teleparallel equivalence with general relativity, and unitary motor causality. We connect Anti-Torsion Fields, helicity-controlled gravity engineering, layer stability, and an experimental roadmap. This paper is explicitly *hypothetical*: macroscopic dual-phase coherence remains unverified.

1 Introduction

Conventional warp-drive proposals, beginning with Alcubierre's metric bubble [1], require exotic negative energy density and violate the null energy condition (NEC) [2]. Engineering such configurations at macroscopic scales appears prohibitive under known physics.

Dual spacetime theory (DST) offers a fundamentally different ontology [3]. There is no autonomous spacetime continuum; instead, *space emerges as collective resonance* among dual rotors carried by individual particles. Gravity is torsional mismatch between the usual and dual sectors, reformulated as teleparallel gravity (TEGR) [6] with vanishing curvature and non-zero torsion. The PGA extension [4] embeds this structure in projective geometric algebra $G(3,1,1)$, adjoining a null vector e_4 so that translations, light propagation, and torsion are unified in a single ten-dimensional bivector object propagated by a unitary motor.

In this framework, **warp propulsion** is not metric engineering but *resonance tuning*: shifting the collective dual-rotor phase to enter repulsive torsional layers ($J < 0$) where effective displacement exceeds the local light-speed bound, while information remains constrained to the light channel. This paper is a *hypothetical* extension of PGA–DST to interstellar propulsion. We preserve the quantitative energy estimates of the earlier DST warp analysis while grounding them in the PGA algebraic structure, Anti-Torsion Field (AT-field) generation [5], and gravity-engineering mechanisms from the PGA textbook programme.

2 PGA–DST Foundations

2.1 Three-sector structure in G(3,1,1)

Each massive particle carries paired compactified Minkowski structures encoded in $\mathrm{Cl}(3,1) \cong \mathbb{H}\otimes\mathbb{H}$. The PGA lift adjoins e_4 with $e_4^2 = 0$ and $\{e_4, e_\mu\} = 0$, extending the algebra to $G(3,1,1)$ (32 real dimensions).

Vector generators are

$$e_0 = j, \quad e_1 = kI, \quad e_2 = kJ, \quad e_3 = kK,$$

with signature $(-,+,+,+)$. The bivector space has dimension ten and decomposes into three sectors [4]:

Table 1: Ten-dimensional generator structure in PGA–DST

Sector	Generators	Square	Role
Hyperbolic	iI, iJ, iK	$+1$	Usual-spacetime boosts
Cyclic	I, J, K	-1	Dual-spacetime rotations
Null	$N_\mu = e_4 \wedge e_\mu$	0	Translations

The null sector realises the Poincaré translation ideal: $N_\mu N_\nu = 0$ for all μ, ν, and the Lie algebra closes as $\mathfrak{iso}(3,1) = \mathfrak{so}(3,1) \oplus \mathfrak{so}(3,1) \ltimes \mathbb{R}^{3,1}$.

2.2 Rotors, motors, and the duality map

The usual and dual rotors are

$$R_{\text{usual}} = \exp\Big(\sum_{a=1}^{3} \frac{\phi_a}{2}\, i\Gamma_a\Big), \qquad\qquad R_{\text{dual}} = \exp\Big(\sum_{a=1}^{3} \frac{\theta_a}{2}\, \Gamma_a\Big), \tag{1}$$

where $\Gamma_a \in \{I, J, K\}$. The relative mismatch rotor is

$$\Omega = R_{\text{usual}}^{\dagger} R_{\text{dual}}, \qquad \Omega_{\text{biv}} = \log \Omega. \tag{2}$$

The duality map $X \mapsto Xi$ (with $i = e_0 e_1 e_2 e_3$) interchanges hyperbolic and cyclic sectors while leaving the null sector closed, since e_4 commutes with i.

The extended mismatch bivector and motor are

$$\Omega^{(5)}_{\text{biv}} = \underbrace{\sum_{a=1}^{3} \Big(\tfrac{\alpha_a}{2} B_a^{+} + \tfrac{\beta_a}{2} B_a^{-} \Big)}_{\Omega_{\text{torsion}}} + \underbrace{\sum_{\mu=0}^{3} \tfrac{\lambda^\mu}{2} N_\mu}_{\Omega_{\text{trans}}}, \tag{3}$$

$$M = \exp(\Omega^{(5)}_{\text{biv}}) = RT, \qquad T = 1 + \sum_{\mu=0}^{3} \tfrac{\lambda^\mu}{2} N_\mu, \tag{4}$$

where the null exponential truncates at first order because $N_\mu N_\nu = 0$.

Theorem 2.1 (Motor unitarity)**.** *Every bivector generator is reversal-odd ($\widetilde{B} = -B$). Consequently $R\widetilde{R} = 1$, $T\widetilde{T} = 1$, and $M\widetilde{M} = 1$. The sandwich product $X' = MX\widetilde{M}$ preserves the degenerate $G(3,1,1)$ metric and causal structure.*

2.3 Torsion scalar and gravitational action

The torsion scalar is defined via the Cartan–Killing form on the six-dimensional torsion sector:

$$J = \tfrac{1}{16}\, B_{\text{Killing}}(\Omega_{\text{biv}}, \Omega_{\text{biv}}) = \tfrac{1}{2} \sum_{a=1}^{3} (\alpha_a^2 - \beta_a^2). \tag{5}$$

The hyperbolic generators contribute $B(B_a^{+}) = +8$ and the cyclic generators $B(B_a^{-}) = -8$, so J changes sign when dual-sector rotation dominates. In the small-angle regime, with $\alpha_a \sim \sinh(\phi_a/2)$ and $\beta_a \sim \sin(\theta_a/2)$, (5) reduces to the finite-angle form

$$J \approx \sum_{a=1}^{3} \Big[\sinh^2\!\Big(\tfrac{\phi_a}{2}\Big) - \sin^2\!\Big(\tfrac{\theta_a}{2}\Big) \Big], \tag{6}$$

used in the operational estimates below.

The PGA extended invariant couples torsion to translation:

$$J^{(5)} = J + \tfrac{1}{2}\eta_{\mu\nu}\lambda^{\mu}\lambda^{\nu}, \tag{7}$$

where $\eta_{\mu\nu} = \mathrm{diag}(-1, +1, +1, +1)$. The translational term vanishes on the light cone $\eta_{\mu\nu}\lambda^{\mu}\lambda^{\nu} = 0$, embedding null propagation algebraically.

Gravity follows from the action

$$S = \frac{c^4}{16\pi G}\int J\sqrt{-g}\,d^4x, \tag{8}$$

which is equivalent to the Einstein–Hilbert and TEGR actions: all GR solutions, including Schwarzschild, are recovered from tetrads generated by unitary motors, without invoking curvature [4, 6]. Interpretation changes; equations of motion do not.

3 What Warp Means in PGA–DST

3.1 Space as collective resonance

In PGA–DST, macroscopic space is not a pre-existing manifold but the statistical alignment of particle-intrinsic rotors across $\sim 10^{32}$ baryons per tonne of matter. The scalar J measures misalignment. When neighbouring dual rotors are aligned ($\theta_a \approx 2n\pi$), $J > 0$ and we experience attractive gravity. When driven to repulsive points ($\theta_a \approx (2n+1)\pi$), collective resonance shifts and effective geometry changes.

Warp propulsion takes two distinct operational meanings:

1. **Resonance-shifting warp (layer-hopping).** The vessel remains within the collective resonance manifold but shifts local dual-rotor phase to enter a repulsive layer ($J < 0$). The spacecraft rides a different harmonic of the same resonant structure. This is the primary mechanism analysed below.
2. **Subspace-transition warp (dual-sector immersion).** The vessel actively decouples from ambient collective resonance and immerses primarily in the dual sector. Re-emergence requires precise re-synchronisation with the target region's phase. This is energetically far costlier than layer-hopping because it breaks macroscopic coherence entirely.

Thus warp is *changing the tuning of cosmic resonance*, not bending or tearing a fabric. The engineering challenge shifts from exotic negative energy to coherent dual-phase control at macroscopic scales.

3.2 Comparison with Alcubierre warp drives

Table 2: Alcubierre metric engineering vs. PGA–DST resonance warp (hypothetical)

Aspect	Alcubierre [1]	PGA–DST (this work)
Mechanism	Contracting/expanding spacetime bubble via $g_{\mu\nu}$	Collective dual-rotor phase tuning; $J < 0$ layers
Exotic matter	Required; violates NEC [2]	Not required; J sign from Killing-form asymmetry
Causality	Local c exceeded inside bubble	Effective superluminal *coast*; local c preserved via $M\widetilde{M} = 1$
Information	Can outrun light within bubble	Bound to light cone; $\eta_{\mu\nu}\lambda^{\mu}\lambda^{\nu} = 0$
GR equivalence	Modified Einstein equations	TEGR/EH recovered; same weak-field limits
Engineering	Negative energy density at bubble wall	THz circularly polarised driving of θ_a

The contrast is ontological as well as operational. Alcubierre engineering deforms a background metric. PGA–DST has no background continuum; it modulates the torsion scalar J and, during coast, the translational component of $J^{(5)}$ on approximately light-like worldlines.

4 Repulsive Layers and Anti-Torsion Fields

4.1 Sign structure of J

Table 3: Torsion scalar sign and physical interpretation

Condition	Sign of J	Effect
$\alpha_a \gg \beta_a$ (aligned)	$J > 0$	Attractive gravity
$\alpha_a \approx \beta_a$	$J \approx 0$	Vacuum / shielding / inertia reduction
$\beta_a > \alpha_a$ (repulsive point)	$J < 0$	Repulsive gravity

Repulsive layers occur near $\theta_a = (2n+1)\pi$. In strong-field astrophysical contexts, the same sign oscillation produces stratified "onion" structures (twist stars) rather than event horizons and singularities [3]: an infinite stack of alternating $J > 0$ and $J < 0$ shells provides the cosmological backdrop for layer-hopping.

Definition 4.1 (Anti-Torsion Field [5])**.** An Anti-Torsion Field (AT-field) is a region $\mathcal{R}$ where $J(x) < 0$. Dual-sector rotation dominates, yielding effective repulsion. AT-fields are engineerable in principle via coherent excitation of θ_a.

For warp entry, an AT-field is not merely a barrier but the *operational medium*: the vessel transitions into a locally generated or cosmically encountered $J < 0$ shell, coasts with reduced gravitational coupling, and exits at a scheduled phase boundary. Stability of AT-fields under perturbation follows from self-reinforcing mismatch dynamics [5]: intrusion of external matter increases $|\beta_a|$, driving J more negative.

5 Drive Mechanism: THz Helicity and Dual-Rotor Coupling

5.1 Electromagnetic splitting

The electromagnetic field splits across DST sectors:

$$F = F_{\text{usual}} + F_{\text{dual}}, \qquad F_{\text{usual}} = \mathbf{E} \cdot (iI, iJ, iK), \qquad F_{\text{dual}} = \mathbf{B} \cdot (I, J, K). \tag{9}$$

Electric fields couple to hyperbolic (usual) generators; magnetic fields couple exclusively to cyclic (dual) generators. This algebraic separation is the geometric origin of the distinct transformation properties under time reversal ($\mathbf{E} \to \mathbf{E}$, $\mathbf{B} \to -\mathbf{B}$).

5.2 Circular polarisation as a chiral switch

For a circularly polarised wave with helicity $\sigma = \pm 1$ propagating along $\hat{z}$,

$$\langle F_{\text{dual}} \rangle \propto \sigma \, E_0^2 \, K. \tag{10}$$

Helicity directly controls J:

- $\sigma = +1$ (right-handed): enhances attractive torsion ($J > 0$).
- $\sigma = -1$ (left-handed): reduces J; sufficient intensity can invert to $J < 0$.

The resonance driving equation is

$$\dot{\theta} = \kappa \, \sigma \, E_0^2 \sin(\omega t - \theta + \delta_\sigma), \tag{11}$$

where κ is a matter-dependent coupling constant. Synchronising θ with ϕ drives $J \to 0$ (shielding, inertia reduction); driving θ past ϕ yields $J < 0$ (repulsion, warp entry).

5.3 THz collective resonance

Single-particle dual resonance lies at Planckian frequencies ($\sim 10^{43}$ Hz). Macroscopic coherence over $N \sim 10^{26}$–10^{32} particles redshifts the effective resonance to the THz band:

$$\omega_{\text{eff}} \sim \frac{\omega_{\text{Planck}}}{\sqrt{N}} \sim 10^{12}\text{–}10^{15}\,\text{Hz}. \tag{12}$$

Superconducting phase locking and high-Q cavities further enhance coupling. This is the same mechanism proposed for table-top tests of $\Delta m/m \gtrsim 10^{-6}$ with helicity-reversal signatures [3]. Warp engineering is the macroscopic extrapolation of that laboratory programme.

6 Operational Regimes: On-Demand Entry vs Layer-Hopping

6.1 Cosmic phase drift

Large-scale structure induces slow background drift of dual-rotor phases:

$$\left|\frac{d\theta_a}{dl}\right| \simeq 1.1 \times 10^{-26}\ \text{rad}\cdot\text{m}^{-1}, \qquad \frac{d\theta_a}{dt} \simeq 3.3 \times 10^{-10}\,(v/c)\ \text{rad}\cdot\text{s}^{-1}. \tag{13}$$

This continuously sweeps the infinite stack of repulsive layers past an observer moving at velocity v.

6.2 Effective Hamiltonian and entry energy

Near a repulsive layer, with $\theta = \theta_a - (2n+1)\pi$,

$$H = \frac{p^2}{2m_{\text{eff}}} + V(\theta), \qquad V(\theta) \approx \tfrac{1}{2}k\theta^2 - \tfrac{\hbar\omega_0}{2}\cos(\theta), \tag{14}$$

where m_{eff} is the collective dual excitation inertia and $\omega_0 \sim 10^{14}$–10^{15} Hz for THz collective modes. The semiclassical energy to reach phase offset $\Delta\phi$ is

$$E(\Delta\phi) \approx N_b\,\hbar\omega_{\text{res}}\left|\sin\!\left(\tfrac{\Delta\phi}{2}\right)\right|, \tag{15}$$

with $N_b \approx 6\times 10^{32}$ baryons in a 10^6 kg vessel. Small $\Delta\phi$ gives quadratic scaling; near π the cost saturates, explaining the extreme sensitivity of on-demand entry.

During entry, the motor (4) is driven primarily through Ω_{torsion} toward $J < 0$. During coast in a repulsive layer, translational mismatch λ^μ is tuned to lie near the light cone so that the $J^{(5)}$ translational term is suppressed and displacement proceeds via phase-layer sliding rather than acausal local boosts.

6.3 On-demand entry energy

Worst-case on-demand entry ($\Delta\phi_{\max} \approx \pi$) for a 10^6 kg vessel:

Table 4: On-demand subspace entry (worst-case $\Delta\phi \simeq \pi$)

Cruise velocity	γ	Energy required (10^6 kg vessel)
$0.10c$	1.005	$\sim 8 \times 10^{11}$ J ≈ 220 GWh
$0.30c$	1.048	$\sim 7.5 \times 10^{11}$ J ≈ 210 GWh
$0.50c$	1.155	$\sim 6.9 \times 10^{11}$ J ≈ 190 GWh
$0.99c$	~ 7	$\sim 3 \times 10^{11}$ J ≈ 80 GWh

These values approach projected compact fusion or antimatter-catalysed limits, making pure on-demand entry technologically challenging.

Table 5: Layer-hopping performance at $v = 0.3c$ for a 10^6 kg vessel

Threshold $\Delta\phi$	Typical wait	Energy required	Power (10-min pulse)	Gain
π (on-demand)	0	750 GWh	—	$1\times$
10^{-1}	$\sim$1 yr	75 GWh	450 MW	$\sim 10^4\times$
10^{-2}	$\sim$40 days	7.5 GWh	45 MW	$\sim 10^5\times$
10^{-3}	$\sim$4 days	750 MWh	4.5 MW	$\sim 10^6\times$
10^{-4}	$\sim$10 hrs	75 MWh	450 kW	$\sim 10^7\times$
10^{-5}	$\sim$1 hr	7.5 MWh	45 kW	$\sim 10^8\times$

6.4 Layer-hopping performance

Waiting time for phase deficit below threshold $\Delta\phi_{\text{threshold}}$:

$$\tau \simeq \frac{\Delta\phi_{\text{threshold}}}{|d\theta/dt|} = 3 \times 10^9 \, \Delta\phi_{\text{threshold}} \left(\frac{c}{v}\right) \text{ s.} \tag{16}$$

A threshold $\Delta\phi \approx 10^{-3}$ offers a practical compromise between waiting time and power. For a 10 light-year mission, total energy expenditure drops to 10–100 MWh under the hybrid protocol.

7 Causality, Horizons, and Effective Superluminal Coasting

7.1 Unitary motors and local causality

Theorem 2.1 guarantees that every physical transformation is implemented by a unitary motor $M\widetilde{M} = 1$. No sandwich product can produce acausal local propagation: the light cone $\eta_{\mu\nu}\lambda^\mu\lambda^\nu = 0$ is built into $J^{(5)}$ (7). Photons correspond to full anti-synchronisation ($\phi = \theta$, $\Omega = 1$) with vanishing translational contribution in $J^{(5)}$.

7.2 Effective superluminal displacement

"Effective superluminal coasting" means that the *phase-layer displacement rate* of the vessel exceeds c as measured against the ambient $J > 0$ reference frame, while:

1. local signal velocity remains $\leq c$ in the vessel's instantaneous frame;
2. information about the hop is carried only on the light channel;
3. no closed timelike curves are generated because hops are finite-duration transitions between globally hyperbolic TEGR patches.

This is analogous to a refractive index change in a medium: phase velocity can exceed c without violating relativity, provided group velocity and information velocity do not.

7.3 Horizons and quasi-horizons

PGA–DST excludes true event horizons and central singularities in favour of stratified twist-star interiors [3]. However, *quasi-horizons*—regions where $\int J \, dr$ rises steeply—can impede exit without supplying sufficient torsion energy. Warp navigation must map local phase gradients to avoid trapping in quasi-horizon shells. This is consistent with the layer-stability analysis below.

8 Mission Protocol, Stability, and Navigation

8.1 Hybrid interstellar protocol

1. Cruise at 0.2–0.5c using nuclear or beamed propulsion.
2. Monitor local dual phase via a compact THz resonant cavity coupled to dual degrees of freedom.

3. When $\Delta\phi < 10^{-3}$ rad (every few days to weeks at $0.3c$), apply a low-power, left-handed circularly polarised THz pulse to complete the transition into $J < 0$.
4. Coast in the repulsive layer for weeks to decades, achieving effective superluminal displacement.
5. Exit via complementary pulse (helicity reversal or phase realignment) and repeat, scheduling hops from phase-gradient maps.

8.2 Layer stability

Inside a repulsive layer, small deviations $\delta\theta$ experience restoring forces from $V(\theta)$ in (14). Characteristic oscillation frequency:

$$\omega_{\text{stab}} \approx \sqrt{\frac{k}{m_{\text{eff}}}}. \tag{17}$$

Active feedback via the same THz cavity damps unwanted excitations. AT-field self-reinforcement [5] provides additional stability against external perturbation.

8.3 Navigation in torsional phase space

Navigation requires control of entry/exit points and hop direction. We propose auxiliary low-mass probes to map local θ_a gradients, enabling predictive scheduling analogous to gravitational slingshots but in torsional phase space. Motor factorisation $M = RT$ separates rotational mismatch (layer selection via R) from translational coast (via light-like T).

9 Experimental Roadmap and Open Questions

9.1 Near-term laboratory tests

Before any warp application, the foundational hypothesis—macroscopic dual-phase coherence coupled to EM fields—must be tested:

1. **Table-top gravity engineering.** Superconducting test masses in high-Q THz cavities; monitor $\Delta m/m \gtrsim 10^{-6}$ with helicity-reversal sign change [3].
2. **AT-field generation.** High-intensity circularly polarised THz (10^{12}–10^{15} Hz) to drive $J < 0$ locally; measure repulsive deflection or weight reduction.
3. **Resonance spectroscopy.** Systematic frequency/helicity scans to constrain κ and compactification scale r_{comp}.

9.2 Astronomical and space-based tests

1. Precision torsion balances or atom interferometers to detect cosmic phase drift (13).
2. Satellite baselines to measure large-scale dual-phase gradients across the solar system.
3. Correlation of repulsive-layer signatures with galactic-scale $J < 0$ boundaries in rotation-curve data [3].

9.3 Key open questions

- What is the precise coupling κ between EM fields and dual rotors?
- What is the decoherence time for macroscopic dual-phase coherence?
- Can collective $N \sim 10^{32}$ baryon locking be maintained during a hop pulse?
- How do quasi-horizon shells interact with repeated layer transitions?

10 Discussion and Conclusions

We have upgraded the dual spacetime warp propulsion hypothesis to the PGA–DST framework. The central claims are:

1. Warp is resonance tuning of collective dual rotors, not Alcubierre metric engineering.
2. Repulsive layers ($J < 0$) and AT-fields provide the operational medium for effective superluminal coasting.
3. THz circularly polarised driving with helicity $\sigma = -1$ is the proposed entry mechanism.
4. Layer-hopping exploits cosmic phase drift to reduce entry energy by 4–7 orders of magnitude relative to on-demand entry.
5. Motor unitarity $M\widetilde{M} = 1$ and the light-cone structure of $J^{(5)}$ preserve local causality.

All quantitative estimates inherit the semiclassical model (15) and remain *hypothetical* until macroscopic dual-phase coherence is experimentally confirmed. Failure of table-top tests would constrain κ and r_{comp} without falsifying the algebraic PGA–DST structure, which independently recovers GR via TEGR. Success at laboratory scales would open a path toward engineering repulsive layers and, in principle, interstellar layer-hopping.

References

[1] M. Alcubierre, "The warp drive: hyper-fast travel within general relativity," *Classical and Quantum Gravity* **11**, L73 (1994).

[2] L. H. Ford and T. A. Roman, "Negative energy, wormholes and warp drive," *Scientific American* **281**, 46 (1999); see also arXiv:gr-qc/0001073.

[3] Anonymous, "Gravity as Torsion between Dual Spacetime: A Biquaternionic Reformulation of General Relativity," December 2025 (in preparation).

[4] Anonymous, "$G(3, 1, 1)$ Extension of Double Spacetime Torsional Mismatch: 10-Dimensional Generator Structure and Plane-Based Bracket Products," 2025.

[5] Anonymous, "Anti-Torsion Field: Engineerable Gravitational Repulsion in Dual Spacetime Theory," 2025.

[6] S. Bahamonde *et al.*, "Teleparallel Gravity: A Review," arXiv:2106.13757 (2021).

[7] C. Gunn, "Geometric Algebra for Computer Graphics," Springer (2017).

[8] V. Lengyel, "Projective Geometric Algebra," BCS (2024).

[9] J. M. Selig, "Geometric Fundamentals of Robotics," 2nd ed., Springer (2005).

[10] C. Doran and A. Lasenby, "Geometric Algebra for Physicists," Cambridge University Press (2003).